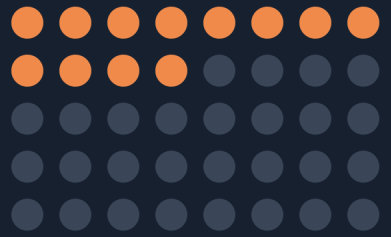
AGENTVERSE 2026 · BIOTECH TRACK



TrialFair

Making drug safety evidence fair to women.

Built for women first, the group that trials have most often left out. Designed to extend to every underrepresented group.

Shafiyyah, Leanne, Weng and Anya

8 of 10

drugs withdrawn from the US market (1997–2000) posed greater health risks to women.

Too few women. 22% of Phase I participants, about 9 per dose group.

Not reported by sex. Results are mostly published as totals.

Costly when found late. After approval: new studies, label changes or withdrawal. Before Phase II: a design decision.

GAO-01-286R (2001) · Labots et al., Br J Clin Pharmacol 2018

Our Multi-Agent Pipeline

Checking whether a completed trial's safety evidence is fair to women, before Phase II

Agent 1

Trial Data Interpreter

- Compares women with men in every dose group
- Side effects, drug exposure and dose response
- Flags where women's data is too thin to tell

Agent 2

Safety Investigator

- Picks the findings that matter most
- Writes labelled hypotheses, not conclusions
- Requests extra analyses or evidence

Agent 3

External Evidence Investigator

- Searches papers and related trials
- Sorts evidence: supports, contradicts, unclear
- Quotes each claim from its source

Agent 4

Trial Improvement Planner

- Turns findings into next steps
- New analyses, data or trial-design changes
- Prioritised for a researcher to approve

Gemini reasons

Chooses tools, interprets text

Python calculates

Every number, deterministically

A researcher decides

Every claim cites a real source

LIVE DEMO

Upload a trial. Watch it investigate.

01

Upload participant-level trial files

02

Follow each agent’s decisions live

03

Check each finding against its source

04

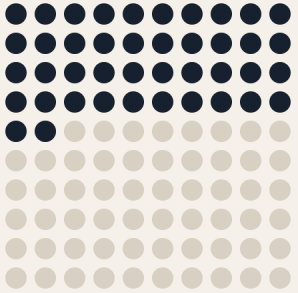
Get suggested changes for the next trial

Does it work? We hid a real difference and checked

We simulated hundreds of trials where women really do get more of the drug, then counted how often each method noticed.

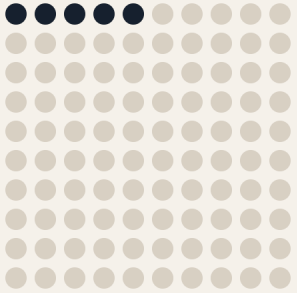
0 in 100

Published results only. They report totals, not women vs men, so the difference is invisible.



42 in 100

TrialFair. Noticed it in 42 trials (71 when the gap is bigger). In the rest it said “too few women to tell”, never “safe”.



5 in 100

False alarms. In trials with no real difference, TrialFair wrongly flagged one only 5 times.

Synthetic trials with about 12 women per dose group, calibrated to 14 approved drugs. Tests TrialFair’s statistics, not real-world accuracy.

Drug safety evidence that is fair to women.

Built for women first. Designed to extend to every group trials underrepresent. Found before Phase II, not after approval.

Safeguards

No invented citations · “not detected” never means “safe” · no dosing advice · researcher has the final say

Next

Other underrepresented groups · sponsors’ own trial data · validation with safety scientists

TrialFair Repo